

communication. This is essential, as mobile has to talk over RF interface to the external world. It gets added during the phone design itself.

The second important component is the Secure Element (SE). It is highly secured crypto memory chip that stores all the information. "Presently, SE is different from a SIM card. But in future, with the standardisation in Single Wire Protocol (SWP), it can be merged with the SIM card," says Dhananjay Dixit, associate manager, NXP Semiconductors.

From the software perspective, an application is written inside a Java J2ME compliant mobile phone. On the other hand, the trusted service manager (TSM) is written/loaded in the secure element where the data is stored. The basic principle it works on is the 'challenge-response' mechanism. In this, when the reader sends the challenge, it is responded in a certain way by the chip on other device. The chip and the reader authenticate each other.

NFC device has the capability to read as well as write without any modification in the existing infrastructure or hardware. "For every application, there is a separate encryption algorithm and a communication protocol that only a dedicated reader can read. So, we can run multiple applications on the same NFC device," states Kandasamy.

"The Secure Element can house multiple secure execution environments. If you have a Citibank credit card application installed on SE, you can use the same SE for transport card or any other card. Multiple applications are separated by application firewalls," Kandasamy adds.

The applications can be tied up with the service operator where over the air (OTA) transactions are required. "It means that data like credit/debit/prepaid/loyalty card, coupon/promotion, product information and others could be provisioned remotely into mobile phones through wireless (3G, 4G) data networks. So, in future, when

A Little World project

A Little World (ALW) had launched its first pilot project for microfinance using NFC technology in India. It used ALW's Zero platform and NFC chip from NXP Semiconductors to convert an NFC mobile phone with large storage capability into a bank branch. It used biometrics for customer identification, mobile phones to open new accounts on the spot, voice prompts for enrolment and transaction and provided printout for each transaction. The breakthrough project was deployed by seven major banks in over 450 villages across four states in India. The project had brought full featured banking services to over 45,000 people right inside their villages.

At present, there are about 8,000 NFC mobile phones in the field and about 200,000 customers satisfactorily using them for financial transactions.

Citibank's 'Tap and Pay'

Citibank has initiated a much larger NFC business trial. Its 'Tap and Pay' programme, launched in Bengaluru in July 2009, has introduced the next generation contactless credit card transactions based on the NFC technology. The technology allows customers to tap their mobile phones on a contactless reader at the point of sale to pay for purchases, eliminating the need for the traditional swiping of credit cards.

The programme is being supported by Nokia 6212 Classic 3G phone, which provides capability to conduct NFC-based transactions. The contactless payment and security infrastructure is being provided by MasterCard PayPass.

VIVOtech is providing NFC wallet, Over-the-Air (OTA) card provisioning software, Smart posters and mobile coupon applications, besides contactless readers to participating merchants.

you buy an NFC phone, you may download your credit or debit card details onto your phone and start making payments through it," says Khan.

Growth avenues and challenges

"When a new technology is introduced, it is very necessary to create an eco-system," says Ashok.

A technology has a certain acceptance curve, and in recent months, NFC has got positive feedback across the globe.

Inter-operability of the connection is guaranteed if the devices comply to the set standards. The technology can resolve the inter-operability issue, but there has to be a union in commercial and business logic suggested by parties sharing the same device.

The service operators are still trying to finalise the business issues to enable NFC support infrastructure and to get millions of NFC phones into the market.

Retrofitting in existing phones is another technological challenge. As NFC chip goes inside the mobile phone design, there is no retrofitting arrange-

ment available. One needs to change a non-NFC mobile handset to the NFC-enabled one to avail the functionality of NFC.

Considering that NFC implementation requires the whole ecosystem to be touched/updated (disrupted), it is taking a longer time than any other technology that does not shake up the whole ecosystem.

"But good news is that the desired types of changes to the whole ecosystem are taking place worldwide that will take the payment, retail and transit industries to the next level. They will offer much more personalised and automated services to consumers than ever before," says Khan.

By 2011, nearly 450 million mobile handsets will be NFC-enabled, states an ABI Research's study. Strategy Analytics forecasts that the mobile phone-based contactless payments will facilitate over \$36 billion consumer spending worldwide by 2011. These are significant projections that will change the way consumers live, work and pay. ●

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